BASIC COMPUTER ENGINEERING (BT-205)

UNIT-WISE MOST IMPORTANT TOPICS (PYQs BASED)

UNIT–I: Computer Fundamentals & Operating System

Top 5 Most Important Topics

1. **Classification of Computers (size & functionality)**

   Asked in: **2019, 2022, 2023, 2025**

2. **Computer Organization (CPU architecture + block diagram)**

   Asked in: **2019, 2024, 2025**

3. **Memory & Storage (RAM, ROM, Cache, Registers, Hard Disk)**

   Asked in: **2019, 2020, 2022, 2023**

4. **Operating System: definition, functions & types**

   Asked in: **2019, 2020, 2022, 2023, 2025**

5. **Applications of Computers (E-business, Health care, GIS, Remote sensing)**

   Asked in: **2020, 2023**

📝 **Pattern:** Pure theory + block diagrams

UNIT–II: Programming Fundamentals & C++ Basics

Top 5 Most Important Topics

1. **Algorithms (definition, need, complexity) + Flowchart**

   Asked in: **2022, 2023, 2024, 2025**

2. **Procedure-oriented vs Object-oriented Programming**

   Asked in: **2019, 2023, 2024**

3. **C++ Tokens, Data types, Variables, Operators**

   Asked in: **2023, 2024**

4. **Control Structures & Looping Statements in C++**

   Asked in: **2019, 2022**

5. **Arrays & Functions (with simple programs)**

   Asked in: **2019, 2022, 2024**

📝 **Pattern:** Theory + small programs

UNIT–III: OOP & Data Structures

Top 5 Most Important Topics

1. **OOP Concepts (Object, Class, Encapsulation, Polymorphism)**

   Asked in: **2019, 2022, 2023, 2025**

2. **Constructors & Destructors (with program)**

   Asked in: **2020, 2022, 2024**

3. **Inheritance (types + example)**

   Asked in: **2019, 2022, 2023, 2025**

4. **Friend Function (concept + program)**

   Asked in: **2020, 2022, 2025**

5. **Function & Operator Overloading**

   Asked in: **2022, 2023, 2024**

📝 **Pattern:** Explain + C++ example

UNIT–IV: Computer Networks & Cyber Security

Top 5 Most Important Topics

1. **OSI Model (layers & functions) – with diagram**

   Asked in: **2019, 2020, 2022, 2024**

2. **TCP/IP Model & OSI vs TCP/IP comparison**

   Asked in: **2019, 2022, 2023, 2025**

3. **Cyber Attacks (DoS, Phishing, Spoofing, Logic Bombs)**

   Asked in: **2019, 2022, 2023, 2024**

4. **Firewall – working principle**

   Asked in: **2019, 2022, 2025**

5. **Good Computer Security Habits / Cyber Laws**

   Asked in: **2020, 2022, 2023**

📝 **Pattern:** Long theory answers

UNIT–V: DBMS & Cloud Computing

Top 5 Most Important Topics

1. **DBMS vs File System & DBMS Architecture**

   Asked in: **2019, 2023, 2024**

2. **DDL & DML Commands (with examples)**

   Asked in: **2019, 2020, 2022, 2024**

3. **Data Independence & Data Dictionary**

   Asked in: **2019, 2023, 2024**

4. **Cloud Computing (definition, models: IaaS, PaaS, SaaS)**

   Asked in: **2019, 2022, 2023, 2025**

5. **Types of Cloud (Public, Private, Hybrid) – Pros & Cons**

   Asked in: **2020, 2022, 2024**

📝 **Pattern:** Explain / short notes

PREDICTED MODEL QUESTION PAPER (Set-1)

Time: 3 Hours **Maximum Marks:** 70

Instructions:

1. Attempt **any five** questions.
2. All questions carry **equal marks**.
3. Draw neat diagrams wherever required.
4. Write C++ programs with proper syntax and comments.

Q1 – COMPUTER FUNDAMENTALS (Very High Probability)

a) Classify computers based on **size and functionality**. Compare **supercomputer, mainframe and embedded systems**.

OR

b) Explain the **architecture and working of CPU** with a neat **block diagram**.

Q2 – MEMORY & OPERATING SYSTEM

a) Explain different types of **computer memories – RAM, ROM, Cache, Registers and Hard Disk**.

OR

b) Define **Operating System**. Explain its **functions and types**. Also explain the concept of **kernel**.

Q3 – ALGORITHMS & PROGRAMMING BASICS

a) What is an **algorithm**? Explain the **need of algorithms**. Write an algorithm to find the **largest of three numbers**.

OR

b) Explain **flowchart**. Draw a flowchart to find the **maximum of two numbers**. State the **limitations of flowcharts**.

Q4 – C++ BASICS (Theory + Program)

a) Explain **data types, tokens, variables and operators in C++** with examples.

OR

b) Write a **C++ program to swap two numbers**. Explain the program.

Q5 – OBJECT ORIENTED PROGRAMMING (Sure Question)

a) Explain the **principles of Object-Oriented Programming (OOP)**.

OR

b) Explain **constructors and destructors in C++** with a suitable program.

Q6 – ADVANCED OOP CONCEPTS

a) What is **inheritance**? Explain **different types of inheritance** with suitable examples.

OR

b) Explain **function overloading and operator overloading** with examples.

Q7 – COMPUTER NETWORKS & SECURITY

a) Draw and explain the **ISO-OSI model**. Write the **function of each layer**.

OR

b) Explain **common cyber threats** such as **malware, phishing and Denial of Service (DoS)** attacks. Also explain the **working of firewall**.

Q8 – DBMS & CLOUD COMPUTING (Easy Scoring)

a) Explain **DBMS architecture**. What is **data independence**?

OR

b) What is **cloud computing**? Explain **cloud service models (IaaS, PaaS, SaaS)** and **types of cloud**.

🔥 WHY THIS PAPER IS IMPORTANT

- Covers **almost all repeatedly asked areas**
- **Unit-III (OOP)** & **Unit-IV (Networks)** highly focused
- Includes **sure-shot topics**:
    - CPU block diagram
    - OSI model
    - OOP principles
    - Inheritance / Constructors
    - Cloud computing

PREDICTED MODEL QUESTION PAPER (Set-2)

Time: 3 Hours **Maximum Marks:** 70

Instructions:

1. Attempt **any five** questions.
2. All questions carry **equal marks**.
3. Draw neat diagrams wherever required.
4. Write C++ programs with proper syntax and comments.

Q1 – COMPUTER ORGANIZATION & APPLICATIONS

a) Explain the **organization of a computer system** with the help of a **block diagram**.

OR

b) Explain the **applications of computers** in **E-Business, Health Care and Remote Sensing & GIS**.

Q2 – MEMORY & SOFTWARE

a) Define **computer memory**. Differentiate between **primary memory and secondary memory**.

OR

b) Differentiate between **System Software and Application Software** with suitable examples.

Q3 – PROGRAMMING FUNDAMENTALS

a) What is a **programming language**? Differentiate between **assembly level language and high-level language**.

OR

b) Explain **control structures in C++** (sequence, selection and iteration) with suitable examples.

Q4 – ARRAYS & FUNCTIONS (Theory + Program)

a) What is an **array**? Explain **one-dimensional and two-dimensional arrays** with examples.

OR

b) Write a **C++ program to arrange elements of an array in ascending order**.

Q5 – OBJECT ORIENTED PROGRAMMING (Very Repeated)

a) Define **Object-Oriented Programming**. Write the **characteristics of OOP**.

OR

b) What is a **class and object**? Explain with a suitable **C++ example**.

Q6 – OOP ADVANCED CONCEPTS

a) Explain the concept of **friend function** in C++ with a suitable program.

OR

b) Explain **virtual functions** and their importance in achieving **runtime polymorphism**.

Q7 – COMPUTER NETWORKS & CYBER SECURITY

a) Differentiate between **ISO-OSI model and TCP/IP model**.

OR

b) Explain **Denial of Service (DoS)** attack and **logic bombs**. Write a short note on **good computer security habits**.

Q8 – DBMS & CLOUD COMPUTING

a) Explain **DDL and DML commands** with suitable examples.

OR

b) What is **cloud computing**? Explain **deployment models** (public, private, hybrid) and write **pros and cons of cloud computing**.
